Slide Builder

Slide Builder 4

The guide: what the program does, every function it has, and how it works underneath.

Program 4.0.0 · with the 4.1 and 4.2 changes built so far

16 October 2026

For the team

For the people who make the decks, the product owner and the maintainer. Every picture is a screenshot of the program, taken on the test workbooks of the repository. The cover uses Slide Builder's own wallpaper and glass.

Part 1

What Slide Builder is

A workbook on the shared folder becomes a deck of slides that follow its numbers.

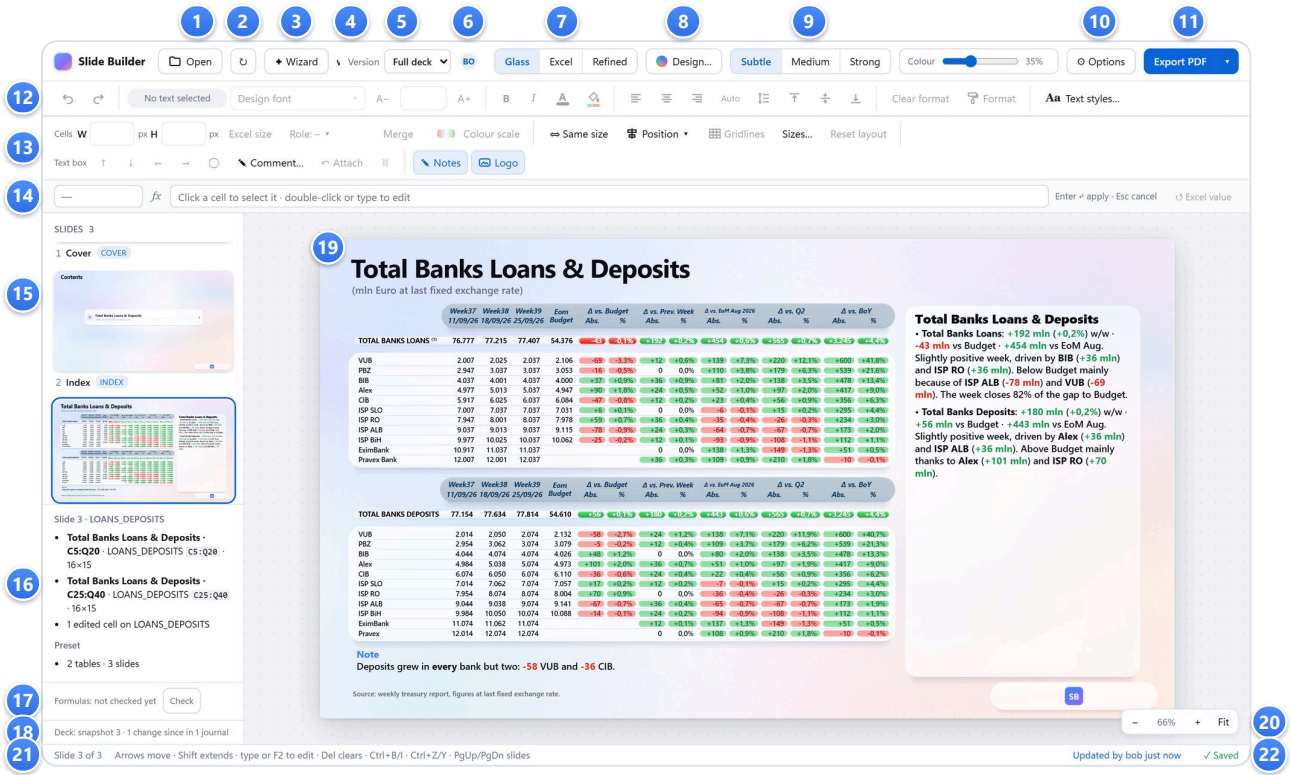
Slide Builder turns tables in an Excel workbook into a deck of 16:9 slides. The workbook stays on the shared folder; Slide Builder reads it and draws the slides from its numbers. When the workbook changes, the slides change with it.

Three designs. Glass (Liquid Glass) redraws each table as translucent glass cards, capsules and fine lines over a coloured background. Excel draws the table exactly as the workbook formats it. Refined (Excel Refined) draws it like Excel and also shows text boxes, automated comments and the notes section. One click switches; each design keeps its own table sizes.

Many people at once. Everybody opens the same workbook from the shared folder and edits the same deck. Other people's changes appear on your screen within 1–2 seconds. Nothing is checked out or locked.

The workbook is never touched. Everything you change – a bold cell, a text box, a colour – is stored beside the workbook, in the folder `backend\data\`. The Excel file itself is never written.

Exports. PDF with real text and lines, PDF made of pictures, PNG pictures, and a copy of the slide for PowerPoint. Several versions and designs can be exported in one go.



1. **Open** – the workbooks of the folder (part 2).
2. ↻ – read the workbook again after it was saved in Excel.
3. ✦ **Wizard** – sheets, tables, slides, versions.
4. The workbook's name and its number of slides.
5. **Version** – which version of the deck you see.
6. Who else has this workbook open.
7. **Glass · Excel · Refined** – the design.
8. **Design...** – colours, text styles, fonts.
9. **Subtle · Medium · Strong** and **Colour** – glass strength and background (Glass only).
10. ⚙ **Options** – page numbers, footer, logo, look.
11. **Export PDF** and its menu ▾.
12. The text toolbar.
13. The layout toolbar: cells, tables, text boxes.
14. The formula bar.
15. The slide list.
16. Notes about the current slide (the Issues panel).
17. The Formulas line (new in 4.2).
18. The deck's health line.
19. The slide, here in Glass.
20. Zoom: –, +, **Fit**.
21. The status bar: the selection and the keys.
22. Who changed what last, and whether your changes are saved.

Part 2

Getting started

The shared folder, starting Slide Builder, opening a workbook, and the wizard's four steps.

The shared folder

All workbooks live in one shared folder, for example `T:\Slide Builder\`. Its parts:

In the folder	What it is
`Start Slide Builder.bat`	starts Slide Builder on your PC (or use the desktop shortcut)
`Install Slide Builder.bat`	the first-time set-up on a PC: checks Python and the folder, makes the desktop shortcut, runs a self-test; nothing is downloaded, no administrator rights needed
`Update Slide Builder.bat`	for the maintainer: installs a new version for everybody (part 8)
your `.xlsx` `.xlsm` `.xlsb` files	the workbooks: read, never changed
`export\`	where PDFs and pictures are saved; `export\feedback\` holds problem reports
`app\<version>\`	the program, one folder per installed version; `app\current.json` says which one runs
`backend\data\`	everything people changed: the decks, shared settings, personal preferences, fonts, logos, backups

Starting

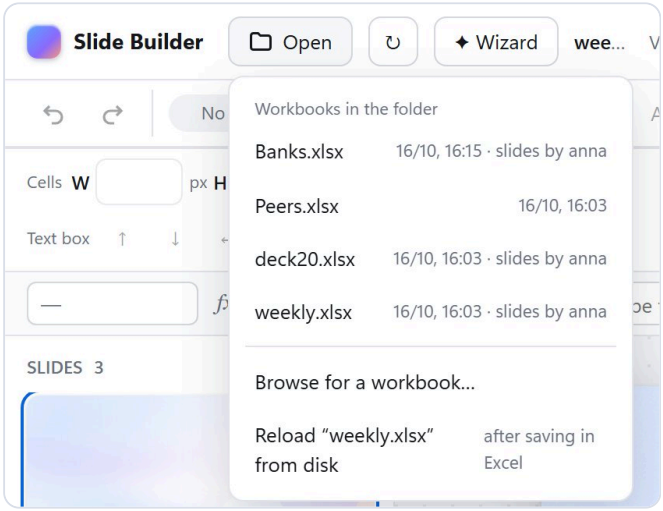
Double-click the desktop shortcut, or `Start Slide Builder.bat`. A black console window opens: keep it open – it is the small program that reads and saves files for you, the **helper**. Slide Builder then opens in its own Edge window, without tabs or address bar, and reopens the last workbook you used.

Opening a workbook

Open lists the workbooks of the folder, newest first, with the time they were saved and "slides by <name>" – the last person who changed that deck. **Browse for a workbook…** picks a file elsewhere; you can also drop a file anywhere on the window. **Reload "…" from disk** (and ↻) reads the workbook again after it was saved in Excel.

`.xlsx`, `.xlsm` and `.xlsb` are read directly, without Excel. An old `.xls`, or a protected file the page cannot read, is converted by the PC's Excel first ("… was converted by Excel for reading – the original file is unchanged"). A file encrypted with a password or a sensitivity label cannot be read, and Slide Builder says so. Hidden sheets are never shown.

A workbook opened for the first time starts the wizard. A workbook whose sheets are named "slide…" and carry "x" tables gets its deck made directly at start-up; the wizard stays available.



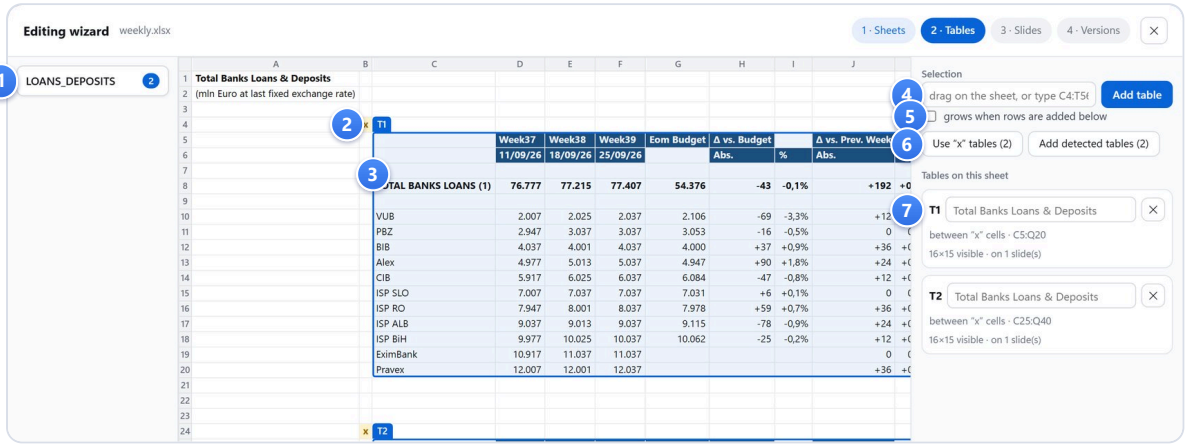
Open. The workbooks of the folder with "slides by anna", Browse, and Reload from disk.

The wizard (✦ Wizard)

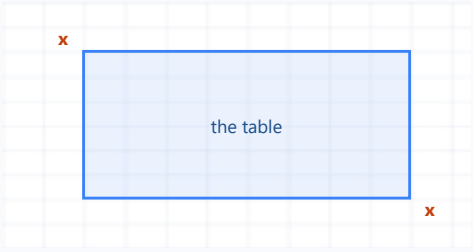
Four steps. The step buttons at the top go back and forth; ✕ or Esc asks before closing. **Save preset & show slides** saves the result – the deck's **preset** – in one change you can undo. If somebody else changed the slides while you were in the wizard, you are asked whether to discard yours or save yours anyway.



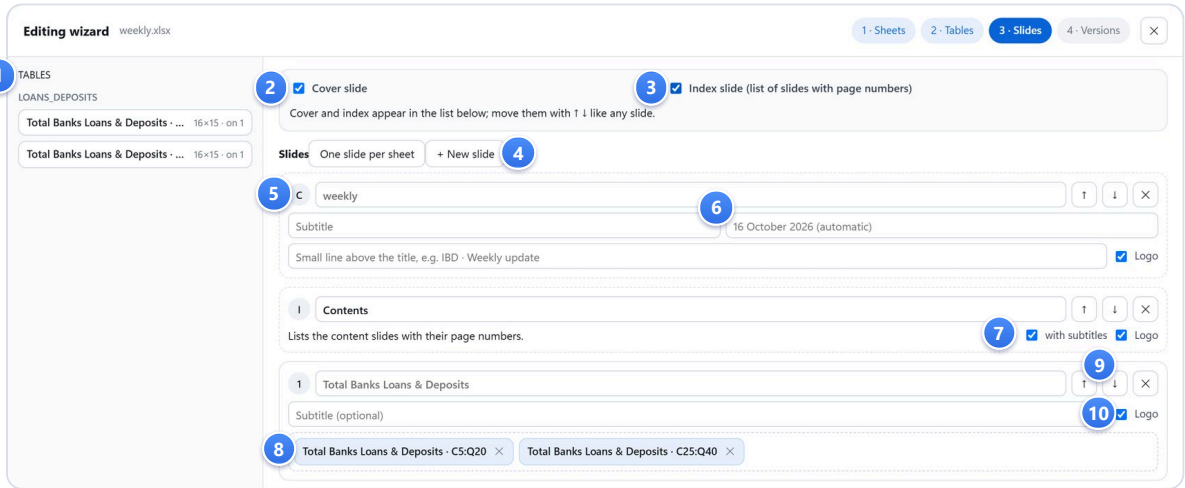
1 · Sheets. Tick the sheets to use; sheets whose name contains "slide" are ticked for you. Each line shows the size, the title and the tables found. **Only "slide" sheets**, **All**, **None**, and **+ Add a linked workbook…** (part 6). In a large workbook only the ticked sheets are read ("not read yet").



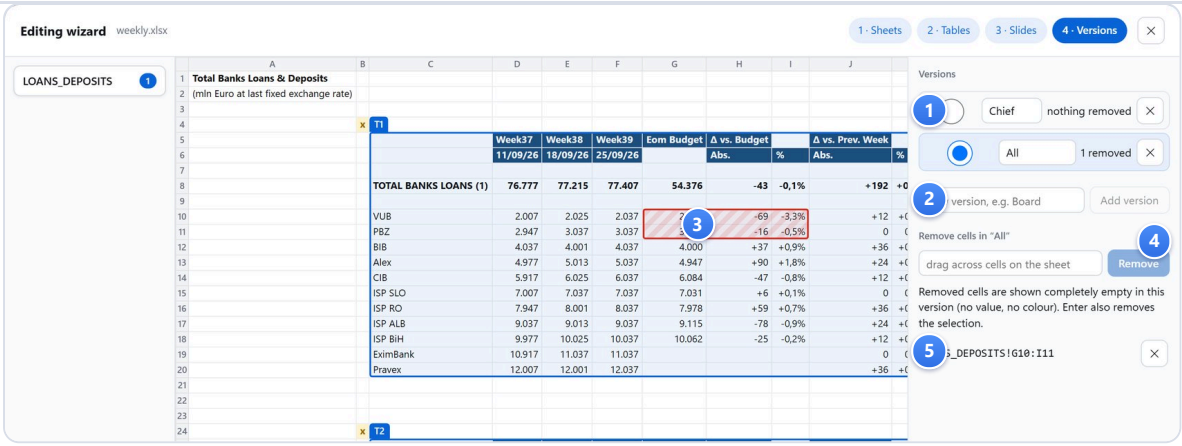
2 · Tables. (1) The chosen sheets, with how many tables each has. (2) An "x" marker. (3) A table: drag across its cells – or Shift+click, Shift+arrows – and press Enter. (4) Or type a range such as `C4:T56` and press **Add table**. (5) **grows when rows are added below** keeps a range table following new rows. (6) **Use "x" tables** takes the marked tables, **Add detected tables** the dashed suggestions. (7) The list names each table, its range and size, and warns about very large ones; ✕ removes one.



"x" markers. Put an `x` in the cell just above and to the left of a table's top-left corner, and another just below and to the right of its bottom-right corner. Slide Builder finds the table between them; the marker row and column are not part of it. The wizard's preview marks them with a yellow "x".



3 · Slides. (1) The tables: drag them onto slides, or click a slide and then the tables. (2) **Cover slide** and (3) **Index slide**, the list of slides with page numbers. (4) **One slide per sheet** and **+ New slide**. (5) The cover's title, subtitle and a small line above the title, (6) its date – "automatic" means today. (7) The index, **with subtitles** or without. (8) The tables of a slide; ✕ takes one off. (9) Move a slide up or down, or remove it. (10) The logo on this slide or not.



4 · Versions. A version is the same deck with some cells removed, for another audience – here *Chief* (everything) and *All* (without the budget figures of two banks). (1) The versions; pick one. (2) Add a version by name, or **+ Chief** / **+ All**. (3) Drag over the cells this version must not show; (4) **Remove** (or Enter). (5) The removed ranges, each with ×. Removed cells are drawn completely empty in that version: no value, no colour.

Part 3

Editing

The slide works like a sheet in Excel. Every change is stored beside the workbook, keyed by sheet and cell.

Click a cell, Shift+click or drag to select more, type to replace the text, F2 or a double-click to edit it, Enter or Tab to move on, Del to clear. Pointing at a cell outlines it. A click on an empty part of the slide clears the selection.

The text toolbar



1. Undo, redo.
2. What it will change: *No text selected*, *Cell C13*, *Cells D10:G12*, *Text box*, *Notes*, *Title*, *Subtitle*, *Cover note*, *Date*.
3. Font: *Design font*, *Font from Excel*, the library, Windows and Google fonts; search, **Upload font files…**, **Manage fonts…**
4. A− / A+ (1 pt steps below 12 pt, 2 pt above) and the size in points.
5. Bold, italic.
6. Text colour: 12 colours, *Automatic*, *Custom…*
7. Cell colour and highlight (below).
8. Left, centre, right, *Auto*.
9. Line and paragraph spacing (text boxes).
10. Top, middle, bottom (text boxes).
11. **Clear format**.
12. **Format**: the format painter.
13. **Aa Text styles…**: every text of a kind, deck-wide.

One toolbar formats any text: cells, a text box, the notes section, or the slide's title, subtitle, cover note and date. Click a title to select it, double-click to edit it on the slide (Enter keeps it, Esc cancels).

Cells

Cell colour

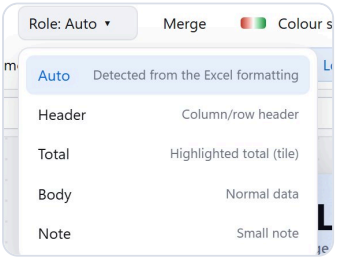
Replaces the fill from Excel: 16 colours, **No colour**, **Colour from Excel**. In Glass it becomes the colour of the block.

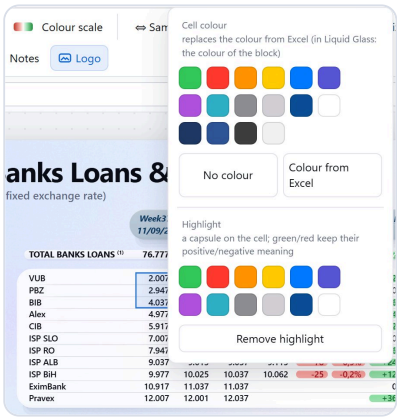
Highlight

A capsule behind the value in Glass; green and red keep their positive / negative meaning. **Remove highlight** takes it away.

Role

How Glass draws the selected cells: *Auto* (detected from the Excel formatting), *Header*, *Total* (a highlighted tile), *Body*, *Note*.





Cell colour and highlight on 12 selected cells.

Merge, Unmerge

Joins the selected cells (the top-left value is kept) or splits merges, also those that come from Excel.

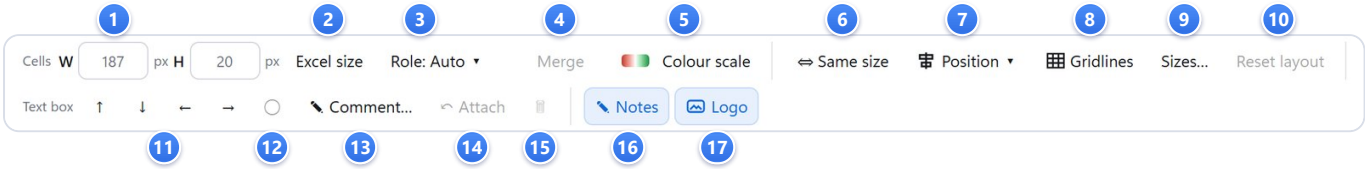
Format painter

Format, then click or drag over cells to copy size, bold, italic, colours, cell colour, highlight, alignment, role, conditional formats and colour scales. A double-click keeps it on; Esc stops. It is greyed until cells are selected.

Formula bar

The name box shows the cell; type in the bar and press Enter (Esc cancels). ↺ **Excel value** goes back to the workbook's own text. A text you typed applies only while Excel's value stays what it was: if the figure changes in the workbook, your text pauses and the Issues panel says so.

Tables



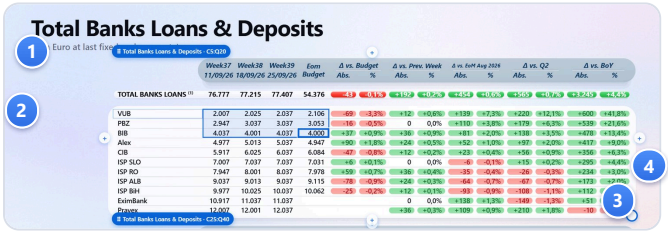
1. W / H of the selected columns and rows, in pixels.
2. **Excel size**: back to the workbook's widths and heights.
3. Role.
4. Unmerge, merge.
5. Colour scale.
6. ⇔ **Same size** for the tables of this slide.
7. Position of the tables.
8. Gridlines.
9. **Sizes...** across slides.
10. **Reset layout** of this slide, in this design.
11. Text box above, below, left, right of the table.
12. Bubble around the selected text box.
13. ✎ **Comment...**
14. ⤺ **Attach** a moved box back to its table.
15. Delete the selected text box.
16. ✎ **Notes**: the notes section.
17. **Logo** on this slide or not.

Sizes on the slide

Type W or H and press Enter, or drag a column or row border on the slide (a guide and a tip show the size); a double-click on the border gives the Excel size. "..." in W or H means the selection has several sizes.

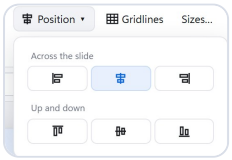
Handles

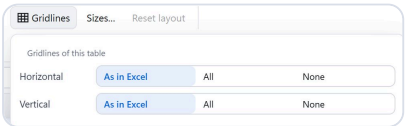
Point at a table and its handles appear: (1) the grip with the table's name – drag it to move the table next to, above or below another; (2) the + buttons add a text box on that side; (3) the corner scales the table; (4) an edge stretches it (0.2–5 ×). Each drag is one change and one undo step.



Position, Gridlines

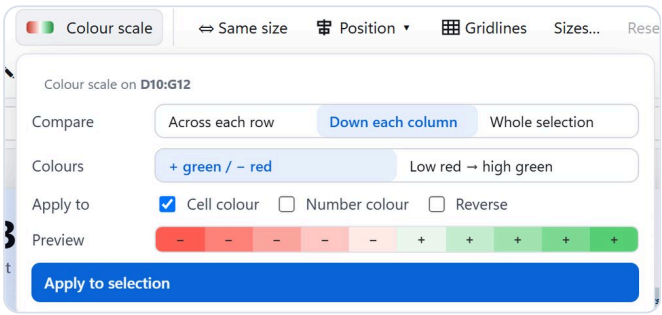
Position places the tables across the slide (left, centre, right) and up and down (top, middle, bottom). **Gridlines** sets the horizontal and vertical lines of the table: *As in Excel*, *All* or *None*.





Colour scale

Deeper green or red the bigger the number: across each row, down each column or over the whole selection; *+ green / – red* or *low red → high green*; on the cell colour or the number colour; *Reverse*; with a preview. **Apply to selection**. The scales of a table are listed in the same menu and removed with ✕.



Sizes...

Pick slides and tables, then **Make same size** (width, height, as the largest, the smallest or the first), copy column widths and row heights from one table to others, align the tables of the chosen slides, or **Reset to Excel sizes**.

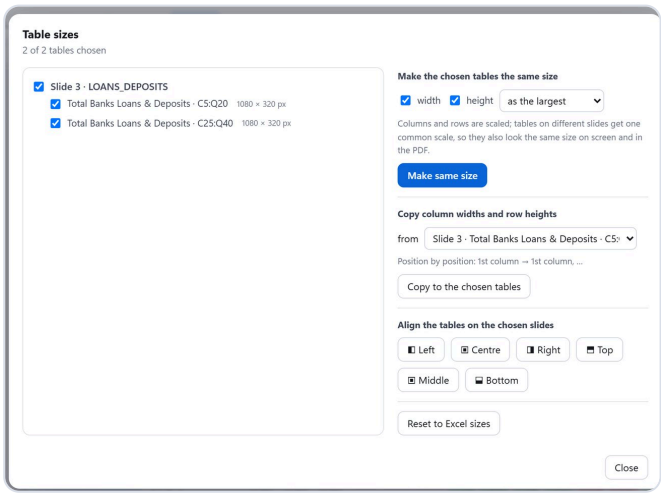
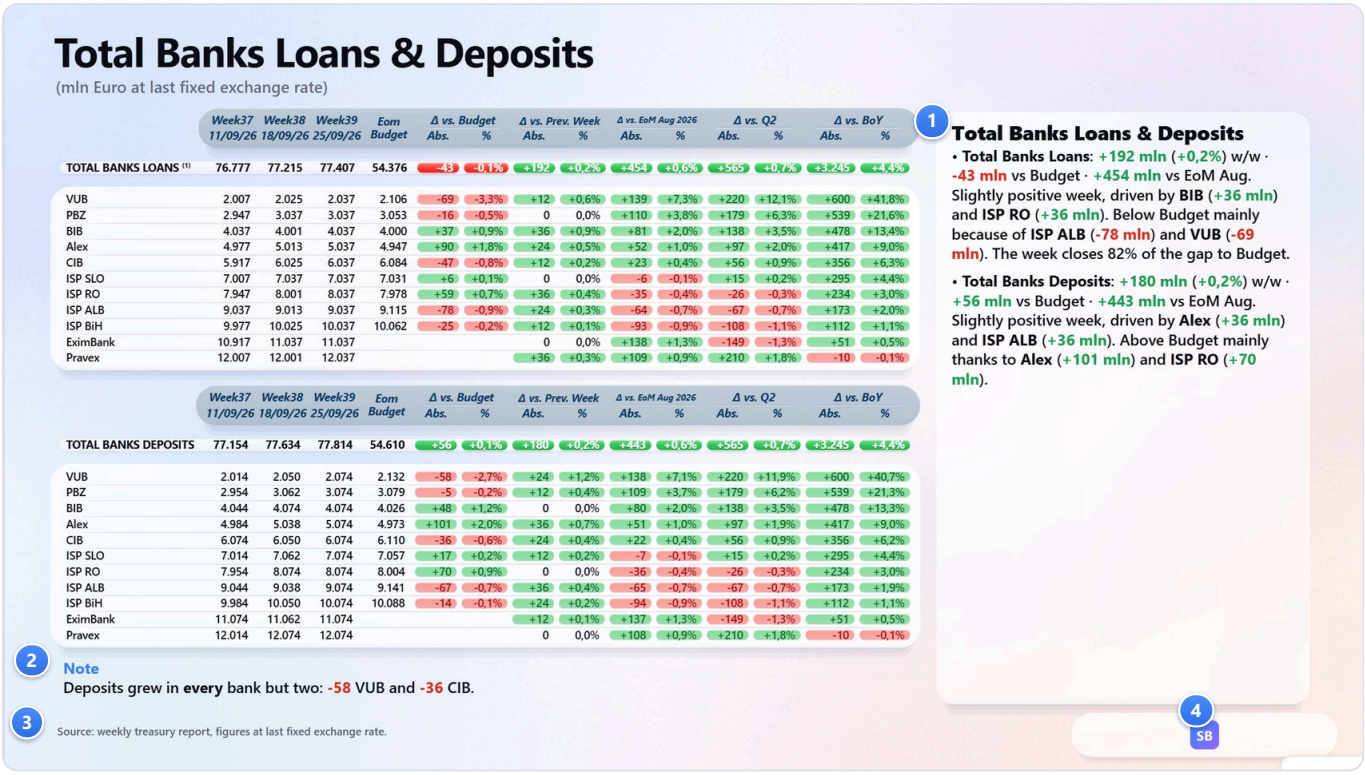


Table sizes belong to the design they were made in: a table can be wide in Glass and narrow in Excel. The tables of a slide are fitted automatically – as large as possible, at most twice their Excel size.

Text boxes and the notes section



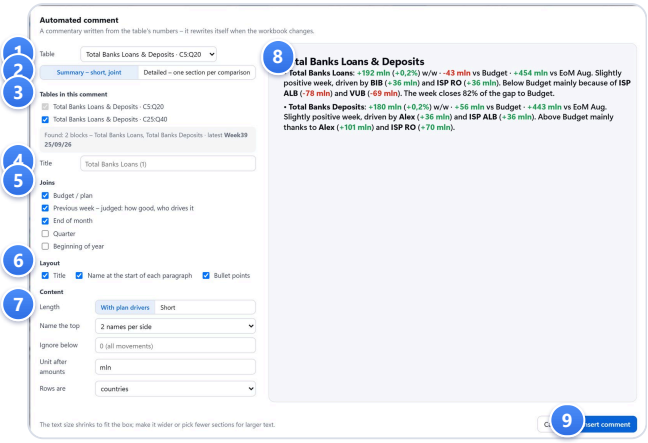
A slide in Glass with (1) an automated comment beside all tables, (2) a text box below the second table using `## Note`, `**every**` and `[[-58]]`, (3) the notes section, (4) the logo in its bubble.

- **Text box** ↑ ↓ ← → adds a box beside the selected table and opens it for writing; the + buttons around a table do the same. Without a selected table it asks you to click a cell first; a side that has a box already says so.
- Markup: `# Title`, `## Heading`, `**bold**`, `[[+12,3]]` (green or red by its sign). Ctrl+Enter or a click elsewhere closes the box, Esc cancels. A new box closed empty disappears.
- A box beside a table is as tall (or as wide) as the table; its text shrinks to fit.
- Click a box to select it; double-click, Enter or F2 edits it; Del deletes it. Drag it anywhere – guides snap it to the tables and the slide, and a moved box takes no room from the tables. Stretch it by its edges.
- ✎ **Notes** adds the notes section: a text box of the slide itself, where the footer goes.
- Text boxes, comments and notes are drawn in Glass and Refined. Excel draws none, and their buttons are greyed there.

Automated comments (✎ Comment…)

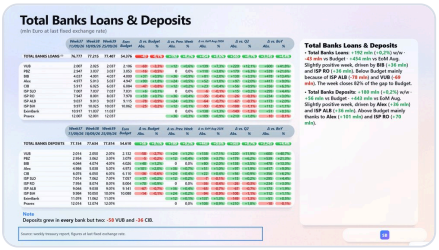
A commentary written from the table's own numbers, like an analyst would. It is rewritten every time the slide is drawn, so it follows new figures, your edits and the version you look at.

- (1) The table, when the slide has several. (2) *Summary* (default): one paragraph per block of the table; *Detailed*: one section per comparison found in the headers (a hint appears when there are none).
- (3) Which tables the comment covers; the "Found:" line says what it recognised: blocks, totals, rows, the latest period.
- (4) A title of your own. (5) Which comparisons to join – those found in the headers, such as Budget, previous week, end of month, quarter, beginning of year. (6) Title, names at the start of each paragraph, bullet points.
- (7) Length; how many names per side (1–5); ignore movements below a value; the unit after amounts (default "mln"); what the rows are (countries, banks, entities…); which rows to include. Further down: where it goes (beside all tables right or left, or above, below, left, right of this table), its width, a bubble.
- (8) The live preview, in the current design. (9) **Insert comment** – later *Update*, or *Remove comment*. *Also on N similar tables* appears when other tables have the same shape.
- Editing the comment's text by hand turns it into fixed text (a toast says so).

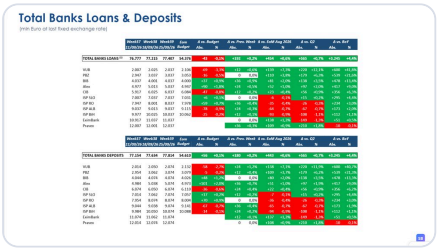


✎ **Comment…** on the weekly table: the summary of loans and deposits.

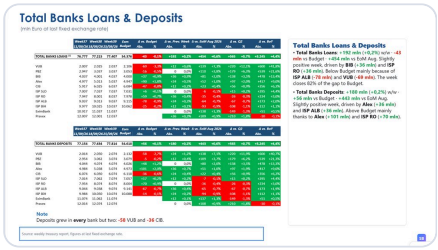
Designs and Design…



Glass – glass cards, capsules, the theme's wallpaper.



Excel – the workbook's own formatting, nothing added.

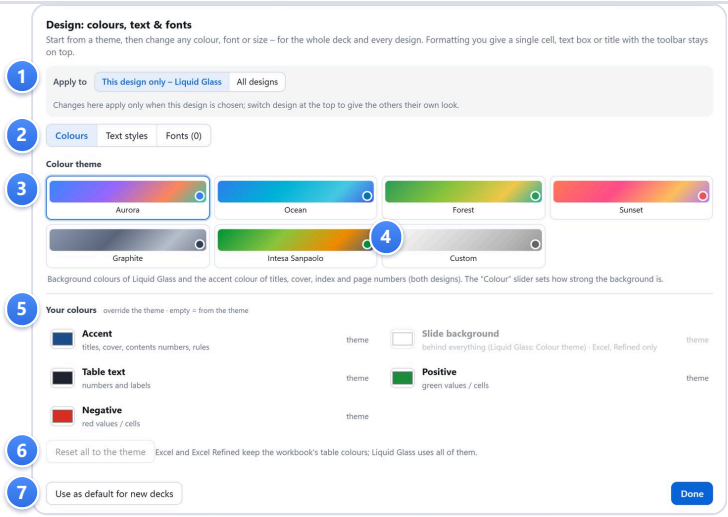


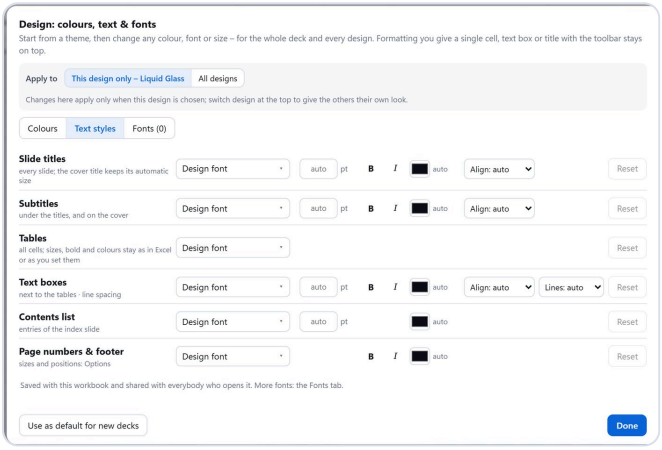
Refined – Excel's tables plus text boxes, comment, notes.

The design is chosen in the top bar and shared with everybody on the workbook. Title, logo, page number and footer stay in every design. Switching design shows the slides with that design's own table sizes. In Glass, **Subtle / Medium / Strong** sets the glass strength and **Colour** (0–100 %) how colourful the background is; moving the slider is one undo step.

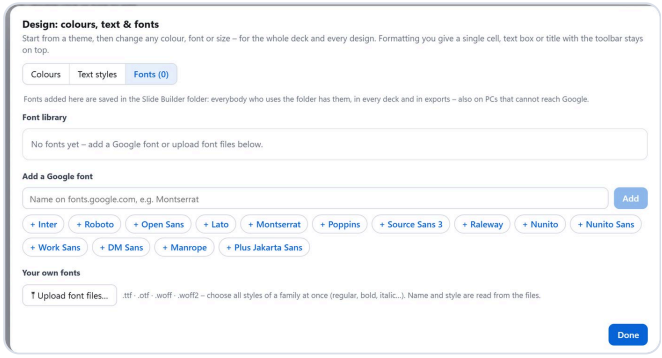
Design… (the button with the theme's swatch) holds the deck's look.

- (1) **Apply to** *This design only* or *All designs*: each design can have its own look.
- (2) Tabs **Colours**, **Text styles**, **Fonts**.
- (3) **Colour theme**: Aurora, Ocean, Forest, Sunset, Graphite, Intesa Sanpaolo, or (4) Custom – four background and two accent colours of your own. The theme colours the Glass background and the accent of titles, cover, index and page numbers.
- (5) **Your colours** override single colours: accent, table text, negative, positive, and the slide background of Excel and Refined; a colour a design does not use is greyed. Each has its own reset; (6) **Reset all to the theme**.
- (7) **Use as default for new decks**: this look becomes the start of every new deck on the folder.



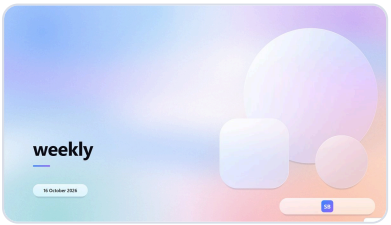


Text styles. Font, size, bold, italic, colour, alignment (and line spacing for text boxes) of all slide titles, subtitles, tables, text boxes, the contents list and page numbers & footer; *Reset* per row. A format given to a single cell, box or title stays on top. A larger title pushes the subtitle and tables down.



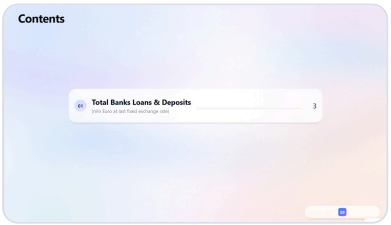
Fonts. The shared font library: add a Google font (downloaded once into the folder, then available to everybody and in exports), suggestions, or upload your own files (.ttf .otf .woff .woff2). The library lists each font with a sample, its styles and who added it, and removes one after a question. It is refreshed about every 30 s.

Cover, index, page numbers, footer, logo



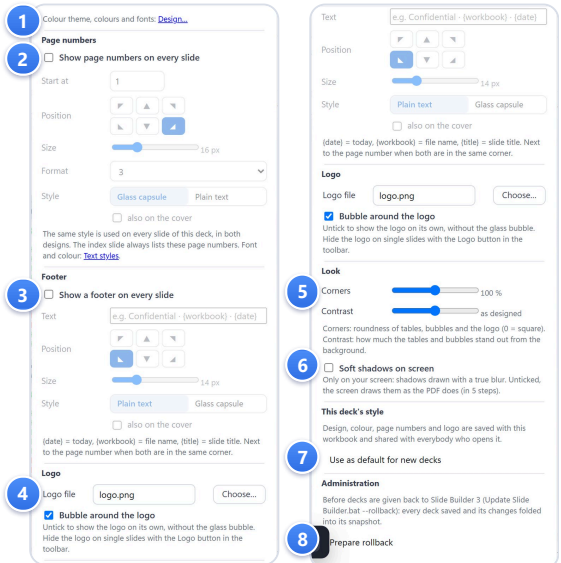
Cover (here at Colour 80 %, Strong): title, date capsule, logo.

Cover and index are made in wizard step 3. Double-click the cover's title, subtitle, note or date to edit them on the slide. The cover's date "automatic" is today's.



Index: every content slide with its page number.

On the index slide the toolbar shows **Subtitles**, which shows or hides the slides' subtitles in the list. The slide list tags these slides *COVER* and *INDEX*.



⚙ **Options** (greyed until a deck is open): (1) a link to Design…; (2) **Page numbers**: on/off, start at, six positions, size, format (*3*, *3 / 12*, *Page 3*, *p. 3*), glass capsule or plain text, also on the cover. (3) **Footer**: a text with `{date}`, `{workbook}`, `{title}`, six positions, size, style, also on the cover; it sits next to the page number when both use one corner. (4) **Logo**: the file – **Choose…** uploads one, copied to the folder for everybody – and a bubble around it; the **Logo** button hides it on single slides. (5) **Look**: corners (0–200 %, the roundness of tables, bubbles and logo) and, in Glass, contrast. (6) **Soft shadows on screen**: a personal choice, a true blur on screen only. (7) **Use as default for new decks**. (8) **Administration** › **Prepare rollback** (part 8). Fonts and colours of page numbers and footer are in Text styles.

Versions

The **Version** list in the top bar shows the deck as *Full deck* or as any version of wizard step 4. Removed cells are empty – no value, colour or highlight, borders kept – and automated comments ignore them. Slide Builder remembers the version you looked at last, per workbook. Exporting every version: part 5.

Undo, the side panel and the status bar

- **Undo / Redo** (↶ ↷, Ctrl+Z, Ctrl+Y or Ctrl+Shift+Z) undo **only your own** changes, up to 200 steps (forgotten when you open another workbook); a toast says what was undone ("Undo: Bold"). A colleague's later change is never undone by you. A new change clears redo.
- **The slide list** shows a picture of every slide, the *COVER* / *INDEX* tags and a yellow badge counting the slide's warnings.
- **The Issues panel** describes the current slide: each table (sheet, range, size, hidden rows, "grows") and what may be wrong (part 8).
- **The deck's health line**: "Deck: snapshot 6 · 3 changes since in 2 journals" (part 7).
- **The status bar** describes the selection ("12 cells selected") and the keys, shows "Updated by <name> just now" and the save state (part 4). Zoom: −, +, **Fit**, 20–300 %, remembered.

Keys	What they do
Arrows, Shift+arrows	move, extend the selection (hidden rows skipped)
Ctrl+A	select the whole table
a letter, F2, double-click	edit the cell; Enter / Shift+Enter, Tab / Shift+Tab move on; Esc cancels
Del, Backspace	clear the cell, or delete the selected text box
Ctrl+B, Ctrl+I · Ctrl+S	bold, italic · save now
Ctrl+Z, Ctrl+Y	undo, redo (your own changes)
PgUp, PgDn · Ctrl+Enter	previous / next slide · close a text box
Esc	stop the painter, close a menu or dialog

Part 4

Working together

Several people edit the same deck at the same time, each on their own PC.

Others' edits appear within about 1–2 seconds

with their name in the status bar: "Updated by bob just now", then "… 3 min ago". If a colleague changes the slides themselves (the preset), your slides are rebuilt and you stay on your slide.

"✓ Saved"

appears as soon as your change is on the shared folder – usually a fraction of a second. Before that: "Unsaved changes…", "Saving…". If the folder does not answer: "⚠ Not saved yet – retrying"; the change is kept and sent again after 2, 4, 8, 16 and 32 seconds. Ctrl+S saves at once. Closing the window with unsaved changes asks first.

Presence

The round initials next to the version list show who else has the workbook open (up to three; on a wide window also "anna is also here"; "(another window)" when it is you in a second window).

Two people, two things

Changes are merged field by field: if you set a column width and a colleague the colour of another cell, both stay.

Two people, the same thing, the same moment

The later change in the deck's own order wins, on every PC alike: everybody ends with the same deck.

Typing is never interrupted

The slide is not redrawn while you write in a cell or box, and settings fields keep what you are typing; the newest slide waits until you finish.

Undo is yours

Undo touches only your own changes, and only the fields you changed, so a colleague's work survives.

Changed on disk

When somebody saves the workbook again in Excel, a banner offers to reload it:

Read-only

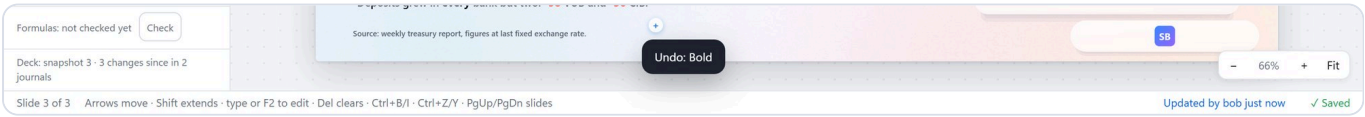
If a deck's file on the share is ever damaged, the deck can still be read but not changed until the maintainer looks at it: a change is refused with "⚠ This deck is read-only", and the health line ends in "· read-only".

Slide Builder was updated

When a newer version that changed the saved formats was installed while you worked, the status bar asks you to reload the page (F5); until then nothing more is saved.

Shared settings

Design, colours, page numbers, footer and logo belong to the deck and are shared. The defaults for new decks are re-read about every 30 s.



The bottom of the window after Ctrl+Z: the Formulas and deck-health lines on the left, the toast "Undo: Bold", and the status bar with "Updated by bob just now" and "✓ Saved".

Part 5

Export

Slide Builder writes PDFs and pictures itself, in the background, from the same drawing you see on screen.

Export PDF exports the whole deck in the PDF mode you chose last (at first: text & tables), with a spinner while it works. The ▾ beside it opens the menu:

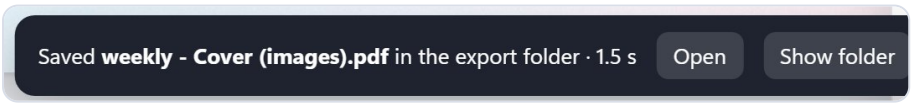
Menu item	What you get
PDF · text & tables	one page per slide, real text and lines, sharp at any zoom – the default
PDF · as pictures	each slide as a 192 dpi picture
PDF · current slide only	the slide on screen
Copy current slide	a 4800 × 2700 picture on the clipboard, to paste into PowerPoint
PNG · current slide	4800 × 2700 picture of the slide
PNG · every slide	one picture per slide
PDF · every version	one PDF per version, in this design
PDF · every version × Liquid Glass + Excel	two PDFs per version
PDF · every version × all three designs	three PDFs per version; a toast counts them ("Saved N PDFs")
Report a problem with this export…	a short report in `export\feedback\` for the maintainer



Export ▾

File names and where they go

Files are saved in the `export\` folder: `<workbook> - slides.pdf`; one slide: `<workbook> - <slide>.pdf`; pictures: `<workbook> - <slide>.png`; versions: `<workbook> - <version> - <design>.pdf`; a PDF of pictures adds " (images)". If a file of that name is open in a viewer, the new one is saved as "… (2)". A toast names the file, with **Open** and **Show folder**:



What to expect

Unsaved changes are saved first, so the export always shows them. The work runs in the background; the window keeps responding. On the 20-slide test deck: a Glass PDF in about 1.4 s and 1.3 MB, an Excel PDF in about 0.4 s and 0.3 MB. A PDF of pictures is much bigger: 11.6 MB in Excel and 69 MB in Glass for the same deck.

Fonts and the page

Each page is exactly one 16:9 slide. The fonts used are embedded, so the PDF looks the same on any PC; a font nobody has is replaced by a default one, and the toast names it. Slide Builder opened straight from a file on disk (an emergency mode without the helper) still exports: the files are downloads instead.

Part 6

New in 4.2

Linked workbooks and the Formulas line – built and checked, not yet released.

Linked workbooks

A deck can draw tables from more than one workbook of the folder. In wizard step 1 press **+ Add a linked workbook…** and pick a file. Its sheets join the list with the file's name beside them; tick them and pick their tables like any other.

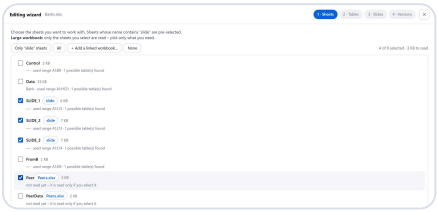
Such a sheet is named `[Peers.xlsx]Peer` on the slides and in the Issues panel: the file in square brackets, then the sheet.

The deck remembers its linked files and opens them with it, read-only like any workbook.

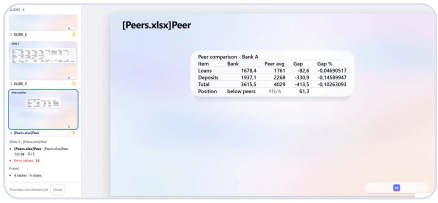
If a linked file is missing when the deck opens, everything else opens and the Issues panel says: "Linked workbook "Peers.xlsx" not found in the folder: its tables are missing until it is back (the deck keeps them)".



+ Add a linked workbook… lists the other workbooks of the folder.



Step 1 of Banks.xlsx with Peers.xlsx linked: its sheets *Peer* and *PeerData* carry the file's name.



A slide made from the linked sheet `[Peers.xlsx]Peer`.

The Formulas line

At the bottom of the side panel, **Check** recomputes every formula of the workbook – and of the folder's workbooks it links to – and compares each result with the value Excel saved in the file. The check changes nothing. The line then reads, for example, "Formulas: 205 of 208 recompute like Excel · 3 locale-dependent (TEXT)"; **Check again** repeats it.

The line says	Meaning, and what to do
recompute like Excel	the result equals Excel's own. Nothing to do.
volatile (TODAY, NOW, RAND)	the value changes every time by nature. Nothing to do.
locale-dependent (TEXT)	a TEXT format code written by an Excel in another language. Nothing to do.
read a hidden sheet · read a workbook not in the folder · could not be read	the check could not judge these cells; put a missing linked file in the folder
different from Excel	shown as a warning: tell the maintainer which workbook it was

The slides always show the values Excel saved in the file. The page is about 2 MB bigger for this: it carries a formula engine that starts only when **Check** is pressed, so opening Slide Builder and a deck is as fast as before.



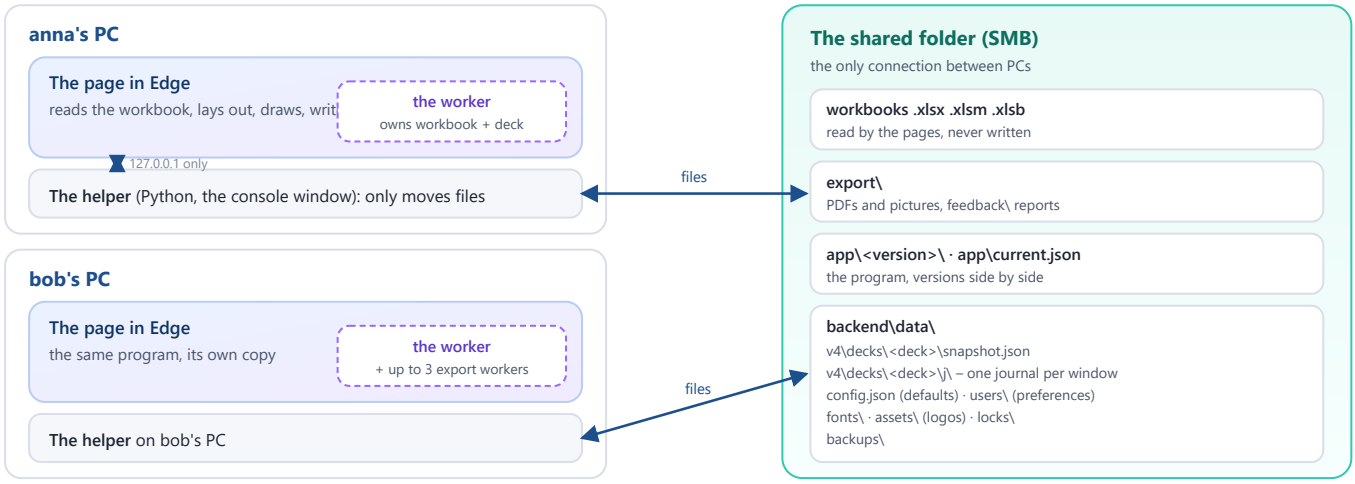
Banks.xlsx after *Check*, with an error value reported above it.

Part 7

How it works underneath

For the owner and the maintainer: what runs where, what is saved where, and why nothing needs a lock.

The pieces



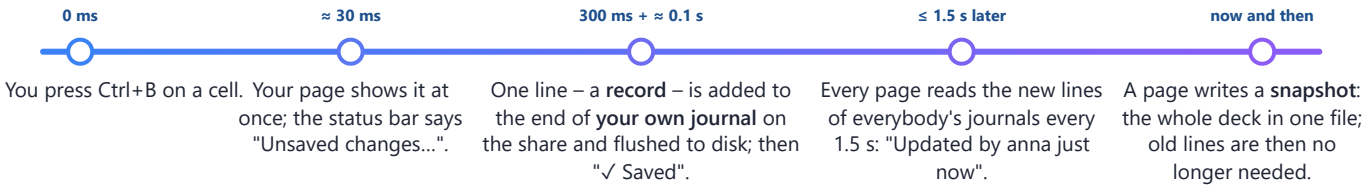
The shared folder holds the workbooks, the exports, the program and all saved state. It is the only connection between PCs: the helpers never talk to each other.

The helper is a small Python program that runs on each PC while Slide Builder is open – the console window. It hands the workbook's bytes to the page, appends changes to journal files, keeps snapshots, saves exports, fonts and logos, and converts old `.xls` files through Excel. It understands none of the slides, and it answers only on the PC itself (127.0.0.1).

The page in Edge does all the work: it reads the workbook, lays out the tables, draws the slides and writes the PDF. It is one file of the program; nothing is installed on the PC beyond Python.

The worker is a background part of the page that owns the open workbook and the deck, so the window never freezes while a big workbook is read. Up to three export workers write PDFs and pictures side by side, and stop after a minute without work.

Saving: journals and snapshots



No lock. Each person (each open window) writes only to their own journal file, so nobody ever waits for anybody. The order of the changes comes from a counter inside each record, not from who reached the share first, so every PC builds exactly the same deck. A record that could not be sent is kept and sent again; the same record is never counted twice.

The health line. "Deck: snapshot 6 · 3 changes since in 2 journals" means: the deck's sixth snapshot, plus 3 changes in 2 journals that are not in a snapshot yet. A snapshot is

Read-only, never guessed. A last line that is still being written is simply read again later. But if a complete line is found damaged, the helper writes an *incident* file in the deck's folder and the deck becomes read-only for everybody until the maintainer looks at it. Nothing is guessed and nothing is lost: an unreadable file is never treated as empty.

Measured. With 20 simulated people on one deck and a slow share (40 ms per file operation), a change was saved in about

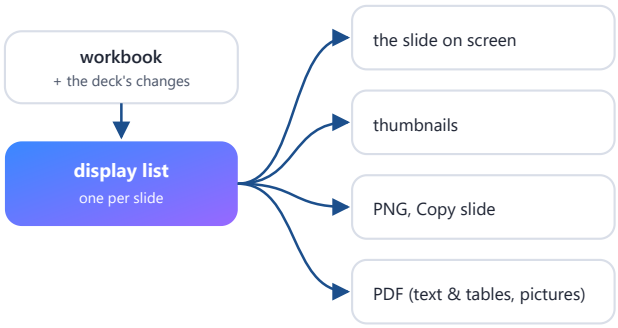
written when more than 2 000 changes piled up, when the oldest is a day old, or when the last person alone on the deck has been quiet for 5 seconds; the previous 30 snapshots stay as backups.

0.1 s and visible to the others in 1.4 s (median), 2.2 s at worst for 95 % of changes. No change was lost in any test.

Drawing: one description for everything

The worker describes each slide once as a **display list**: the rectangles, lines, texts and pictures it is made of, in slide coordinates. The same list draws the slide on screen, the thumbnails, the PNG pictures and the PDF – so what you see is what you export. Only the tables that changed are drawn again after an edit; a key press reaches the screen in about 25–42 ms.

Fonts. Text is measured from the font files themselves, not by the browser, so line breaks and text-box fitting are the same on screen and in the PDF, and on every PC that has the same fonts. Windows fonts can differ by version between PCs; the side panel names such a font ("Calibri on this PC is another version than on PC-B"), whose slides may wrap a little differently there.



The formula engine (4.2)

The formulas are read from the workbook file itself (`.xlsx` and `.xlsb`) and recomputed inside the page by an embedded engine, IronCalc (open source, MIT / Apache-2.0). Excel is not used and not needed. The engine starts only on first use, and today it serves the **self-check** of the Formulas line: every formula cell recomputed and compared with the value Excel saved. The slides keep showing Excel's saved values.

What is never done

- **The workbook is never written.** Slide Builder only reads it; everything people change lives in `backend\data\` .
- **Decks and workbooks go nowhere but the shared folder.** The helper answers only on the PC itself and needs a key the page receives at start. Only two things reach the internet: the font menu loads previews of Google font names (and adding one downloads it), and `Update Slide Builder.bat` downloads a new release from GitHub.
- **Hidden sheets are never shown**, on purpose.

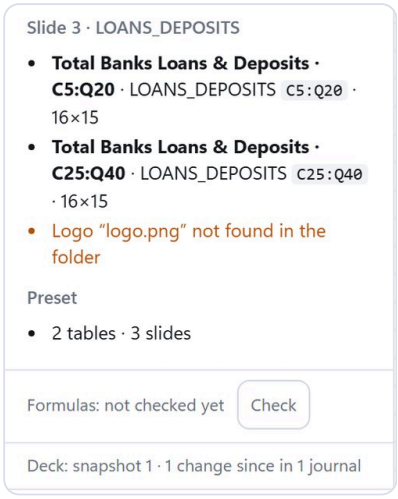
Part 8

When something looks wrong

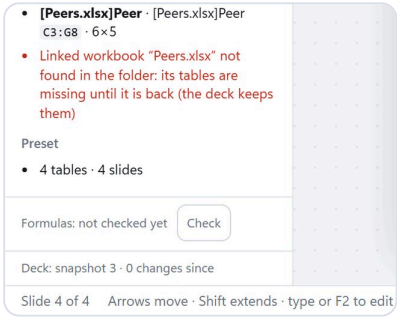
The Issues panel, the common messages, and – for the maintainer – logs, backups, updates and roll-backs.

The Issues panel

The line says	Meaning, and what to do
Error values: D11	the cell shows an Excel error (#N/A, #DIV/0!…): fix it in the workbook
Small text (≈8.6px at 100%)	the table is shrunk a lot: enlarge it or move it to its own slide
N tables of this slide could not be found	a sheet was renamed or deleted: open the wizard
No tables on this slide	add tables in wizard step 3
Chart on … not reproduced	charts are not drawn: in Excel, paste the chart as a picture inside the table area
Pictures that could not be read · conditional formats not reproduced	that element is left out
N edited cells · M text edits paused	Excel's value changed under your text (new figures, rows inserted above): check and retype
Logo "logo.png" not found in the folder	choose the logo again in ⚙ Options
Linked workbook "…" not found in the folder	put the file back in the folder
… on this PC is another version than on …	a Windows font differs between PCs: slides may wrap a little differently there



The tables of a slide and a missing logo.



A linked workbook moved away.

Common messages

Message	What to do
"… is protected (password or sensitivity label with encryption)"	remove the protection, or save an unprotected `.xlsx` copy
"Converting … with Excel…" (an old `.xls`)	wait; this needs Excel on the PC. The original file stays unchanged
"Helper not running" / "Helper stopped"	the console window was closed: start Slide Builder again
"⚠ Not saved yet – retrying"	the share is slow or unreachable; leave the window open, the change is kept
"Slide Builder was updated – reload the page (F5)"	press F5
"⚠ This deck is read-only"	call the maintainer

For the maintainer

Logs and reports

The helper writes its log in its console window, with the time of each line; there is no log file, so copy the lines from the window. An unexpected error shows its details there too. Users' export problem reports are in `export\feedback\`. A damaged journal leaves `incident-….json` in `backend\data\v4\decks\<deck>\`; repairing such a deck is a manual job.

Backups

In `backend\data\backups\`: the last 30 snapshots of every deck (`v4\<deck>\`), each file before a format upgrade (`upgrades\`), and the saved setups before each update (`before-update-<time>\`, the last 5).

Updating

Run `Update Slide Builder.bat`. It downloads the newest release, checks it, installs it beside the current one in `app\<version>\`, runs its self-test on the folder, and only then makes it current. Then everybody closes Slide Builder, starts it again and reloads the page. An update never changes saved decks, settings, workbooks or exports. One update runs at a time.

Rolling back

The program: `Update Slide Builder.bat --use 4.0.0` (or another installed version) makes it current again; after going back, a plain update refuses until you name a version on purpose. **A deck to Slide Builder 3**, while `Start Slide Builder (v3).bat` still exists: ⚙ **Options › Administration › Prepare rollback**, everybody closes the deck, then `Update Slide Builder.bat --rollback "<workbook>"`, and the people on that deck use the v3 start file. Details: `docs/next/cutover.md` §3.

Part 9

Glossary

deck	the slides of one workbook, with everything people changed; stored beside the workbook in `backend\data\`
preset	the deck's setup made in the wizard: sheets, tables, slides, versions
version	the same deck with some cells removed, for another audience (e.g. *Chief*, *All*)
design	how tables are drawn: Glass (Liquid Glass), Excel, Refined (Excel Refined)
theme	the set of colours of a deck: Aurora, Ocean, Forest, Sunset, Graphite, Intesa Sanpaolo or Custom
role	how Glass treats a cell: header, total, body or note (Auto = detected from the Excel formatting)
highlight	a capsule behind a value in Glass
"x" marker	an `x` in the cells just outside two opposite corners of a table, so Slide Builder finds it
helper	the small program in the console window that moves files between the page and the shared folder
journal	the file of changes one open window makes to a deck, added to line by line, never rewritten
snapshot	the whole deck in one file; the journals hold the changes made since
display list	the description of a slide that draws the screen, the pictures and the PDF alike
linked workbook	another workbook of the folder whose sheets a deck uses; its sheets are named `[File.xlsx]Sheet`
scenario	a run of the deck for several values of one input cell – planned for 4.2, not yet in the program
self-check	the Formulas line's comparison of every recomputed formula with the value Excel saved